Lecture 06: Generalized Linear Models

Machine Learning Tutorial

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1. From regression to exponential-family form

Let $z = \mathbf{x}^\top \mathbf{w}$.
Gaussian regression uses $\mu = z$ (variance 1);
logistic regression uses $\mu = \sigma(z)$, where $\sigma(z) = e^z/(1+e^z)$.

$$p_G(y;\mu) = \frac{1}{\sqrt{2\pi}} e^{-(y-\mu)^2/2} = \underbrace{\frac{e^{-y^2/2}}{\sqrt{2\pi}}}_{h(y)} \exp\Big[\underbrace{\mu}_{\eta}\, y - \underbrace{\frac{\mu^2}{2}}_{A(\eta)}\Big], \qquad y \in \mathbb{R},$$

$$p_B(y;\mu) = \mu^y (1-\mu)^{1-y} = \underbrace{1}_{h(y)} \exp\Big[\underbrace{\log\frac{\mu}{1-\mu}}_{\eta}\, y - \underbrace{\log(1+e^\eta)}_{A(\eta)}\Big], \quad y \in \{0,1\},\ 0 < \mu < 1.$$

Both have the form

$$p(y;\eta) = h(y)\exp[\eta y - A(\eta)].$$

The natural parameter η is the coefficient of y; here $\eta = z$ in both models.

2.1. Definition of an exponential family

Definition. Fix an output domain $\mathcal{Y}$ and functions

$$h : \mathcal{Y} \to [0, \infty), \qquad \phi : \mathcal{Y} \to \mathbb{R}^k,$$

independent of η. Define

$$Z(\eta) := \int_{\mathcal{Y}} h(y) e^{\eta^\top \phi(y)} \, dy, \qquad \Theta := \{\eta \in \mathbb{R}^k : 0 < Z(\eta) < \infty\}.$$

Assume $\Theta \neq \varnothing$ and set $A(\eta) = \log Z(\eta) \in \mathbb{R}$ for $\eta \in \Theta$. The probability mass functions or densities

$$p(y; \eta) = h(y) \exp\left[\eta^\top \phi(y) - A(\eta)\right], \qquad y \in \mathcal{Y}, \ \eta \in \Theta,$$

form an **exponential family** over all normalizable natural parameters.

For discrete outputs, replace the integral by $\sum_{y \in \mathcal{Y}}$.

The function h need not be normalized. The fixed $\mathcal{Y}, h, \phi$ determine A.

η: Natural parameter

$\phi(y)$: Sufficient statistic

$h(y)$: Base measure

$A(\eta)$: Log-partition function

References: Stanford CS229, Sec. 8, and W. Fithian, Berkeley STAT 210A.

2.2. Property 1: Convexity of $A(\eta)$

Definition. Fix $\mathcal{Y}$, $h : \mathcal{Y} \to [0, \infty)$ and $\phi : \mathcal{Y} \to \mathbb{R}^k$. The **natural parameter space** is

$$\Theta := \{\eta \in \mathbb{R}^k : 0 < Z(\eta) = \int_{\mathcal{Y}} h(y) e^{\eta^\top \phi(y)}\, dy < \infty\}.$$

Theorem. For the fixed $\mathcal{Y}, h, \phi$ above, if $\Theta \neq \varnothing$, then Θ is convex and A is convex on Θ.

Proof. Take $\eta_0, \eta_1 \in \Theta$, $0 < t < 1$, and $\eta_t = t\eta_0 + (1-t)\eta_1$.

$$Z_i := Z(\eta_i), \qquad p_i(y) := \frac{h(y) e^{\eta_i^\top \phi(y)}}{Z_i}, \qquad \int_{\mathcal{Y}} p_i(y)\, dy = 1 \quad (i = 0, 1).$$

$$\frac{Z(\eta_t)}{Z_0^t Z_1^{1-t}} = \int_{\mathcal{Y}} p_0(y)^t p_1(y)^{1-t}\, dy \le \int_{\mathcal{Y}} \left[t p_0(y) + (1-t) p_1(y)\right] dy = 1.$$

$$\underbrace{0 < Z(\eta_t) \le Z_0^t Z_1^{1-t} < \infty}_{\eta_t \in \Theta:\ \Theta \text{ is convex}}, \qquad \underbrace{A(\eta_t) = \log Z(\eta_t) \le t A(\eta_0) + (1-t) A(\eta_1)}_{A \text{ is convex on } \Theta}.$$

Common support ensures $Z(\eta_t) > 0$. Endpoints $t = 0, 1$ follow directly. Use sums for discrete outputs.

References: W. Fithian, Berkeley STAT 210A, and G. Deligiannidis, Oxford, Sec. 1.3 (Definition 1.12).

Optional: The weighted arithmetic-geometric mean inequality

$$a, b > 0, \qquad 0 < t < 1.$$

$$f(x) = \log x, \qquad \underbrace{f''(x) = -\frac{1}{x^2} < 0 \quad (x > 0)}_{\log \text{ is strictly concave}}.$$

$$\log[ta + (1-t)b] \geq t \log a + (1-t) \log b = \log(a^t b^{1-t}).$$

Since $\exp$ is strictly increasing,

$$\boxed{\underbrace{a^t b^{1-t}}_{\text{weighted geometric mean}} \leq \underbrace{ta + (1-t)b}_{\text{weighted arithmetic mean}}}.$$

Equality for $0 < t < 1 \iff a = b$.

2.3. Property 2: Derivatives of $A(\eta)$ give the mean and covariance

If $\operatorname{int}\Theta \neq \varnothing$, then for $\eta \in \operatorname{int}\Theta$, A is smooth and

$$\nabla A(\eta) = \frac{1}{Z(\eta)} \int_{\mathcal{Y}} \phi(y) h(y) e^{\eta^\top \phi(y)}\, dy = \underbrace{\mathbb{E}_\eta[\phi(Y)]}_{\text{mean of the statistic}},$$

$$\nabla^2 A(\eta) = \mathbb{E}_\eta[\phi(Y)\phi(Y)^\top] - \mathbb{E}_\eta[\phi(Y)]\mathbb{E}_\eta[\phi(Y)]^\top = \underbrace{\operatorname{Cov}_\eta[\phi(Y)]}_{\text{covariance of the statistic}} \succeq 0.$$

3.1. Definition of a Generalized Linear Model (GLM)

For prediction, let the exponential-family parameter η depend on the input $\mathbf{x}$.

Definition (GLM, scalar case). A **generalized linear model** combines

- Distribution: $p(y_n \mid \mathbf{x}_n; \mathbf{w}) = h(y_n) \exp\left[y_n \eta_n - A(\eta_n)\right]$.
- Linear score: $z_n = \mathbf{x}_n^\top \mathbf{w}$.
- Mean and link: $\mu_n := \mathbb{E}[Y_n \mid \mathbf{x}_n], \qquad g(\mu_n) = z_n$.

Here $\phi(y) = y$, $\eta_n \in \operatorname{int} \Theta$, and g is invertible on the model's mean range.

The family and link are chosen; one shared weight vector $\mathbf{w}$ is learned.

$$\underbrace{\mu_n = g^{-1}(\mathbf{x}_n^\top \mathbf{w})}_{\text{predicted mean}}, \qquad \mu_n = A'(\eta_n).$$

The **canonical link** makes the natural parameter equal to the linear score:

$$\eta_n = z_n = \mathbf{x}_n^\top \mathbf{w} \quad \Longrightarrow \quad g^{-1}(z_n) = A'(z_n).$$

For this canonical choice, assume A' is invertible on $\operatorname{int} \Theta$. A general GLM may use another link.

3.2. Negative log-likelihood learns the GLM weights

Given inputs, assume independent outputs. Use the canonical scalar model:

$$\eta_n = z_n = \mathbf{x}_n^\top \mathbf{w}, \qquad \mu_n = A'(z_n).$$

$$\underbrace{J(\mathbf{w}) = -\frac{1}{N}\sum_{n=1}^{N} \log p(y_n \mid \mathbf{x}_n; \mathbf{w})}_{\text{minimize over the shared weights}} = \frac{1}{N}\sum_{n=1}^{N} \left[A(z_n) - y_n z_n - \log h(y_n)\right].$$

$$\nabla_{\mathbf{w}} J = \frac{1}{N}\sum_{n=1}^{N} \left[A'(z_n) - y_n\right] \mathbf{x}_n = \frac{1}{N}\sum_{n=1}^{N} \underbrace{(\mu_n - y_n)\mathbf{x}_n}_{\text{mean minus observation, times input}}.$$

One solver is gradient descent. Recompute the means and gradient at each iteration:

$$\boxed{\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \alpha_t \nabla J(\mathbf{w}^{(t)})}, \qquad \alpha_t > 0.$$

After T iterations, predict $\mu^{(T)}(\mathbf{x}) = A'(\mathbf{x}^\top \mathbf{w}^{(T)})$.

Require $h(y_n) > 0$ and interior scores. Use admissible steps that reduce J when the gradient is nonzero. A finite optimum and convergence are not automatic.

3.3. Specific examples of GLMs

Canonical examples: $\phi(y) = y$, $\eta = z = \mathbf{x}^\top \mathbf{w}$, $\mu = A'(z)$, $\Theta = \mathbb{R}$.

	Least squares	Logistic regression	Poisson regression
Distribution of $Y \mid \mathbf{x}$	$\mathcal{N}(\mu, 1)$	$\text{Bernoulli}(\mu)$	$\text{Poisson}(\mu)$
Output y	$\mathbb{R}$	$\{0, 1\}$	$\{0, 1, 2, \ldots\}$
$h(y)$	$e^{-y^2/2}/\sqrt{2\pi}$	1	$1/y!$
$A(\eta)$	$\eta^2/2$	$\log(1 + e^\eta)$	e^η
Link $g(\mu) = \eta$	μ	$\log[\mu/(1-\mu)]$	$\log \mu$
Mean $\mu = g^{-1}(z) = A'(z)$	z	$\sigma(z)$	e^z
NLL $\ell(y, z)$	$\frac{1}{2}(y-z)^2 + \frac{1}{2}\log(2\pi)$	$\log(1 + e^z) - yz$	$e^z - yz + \log(y!)$

$$\ell(y, z) = -\log p(y \mid \mathbf{x}; \mathbf{w}), \qquad J(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^{N} \ell(y_n, \mathbf{x}_n^\top \mathbf{w}).$$

The family determines the mean and loss; the learning procedure stays the same.

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